MongoDB – Simple CRUD & Group By Practice Sheet

Beginner Level | Practice with MongoDB Compass or mongosh

Learning Objectives

• Create a database and collection.

• Insert documents into a MongoDB collection.

• Perform basic CRUD operations: Create, Read, Update, Delete.

• Use simple comparison and sorting queries.

• Use the aggregation pipeline with $group.

1. Create Database and Collection

Use the following database and collection:

Database: collegeDB
Collection: students

2. Sample Data

Insert the following 8 documents into the **students** collection:

Name	Age	City	Course	Marks
Rahul	20	Mumbai	BCA	78
Priya	21	Pune	BCA	85
Amit	20	Mumbai	BSc	72
Neha	22	Surat	BCA	91
Rohan	21	Pune	BSc	68
Sneha	20	Vadodara	BCA	88
Karan	22	Surat	BSc	75
Anjali	21	Vadodara	BCA	82

3. CRUD Practice Questions

Q1. Create: Insert one new student named **Vivek**, age 22, city Ahmedabad, course BSc, marks 80.

Q2. Read: Display all students.

Q3. Read: Display only students whose course is **BCA**.

Q4. Read: Display students who scored more than **80** marks.

Q5. Read: Display students who live in **Vadodara**.

Q6. Read: Display only the **name** and **marks** fields of all students.

Q7. Update: Change Priya's marks from 85 to **90**.

Q8. Update: Change Rohan's city from Pune to **Mumbai**.

Q9. Delete: Delete the student named **Karan**.

Q10. Delete: Delete all students whose course is **BSc**.

4. Sorting and Limiting

Q11. Display students sorted by marks from highest to lowest.

Q12. Display the top 3 students according to marks.

Q13. Display students sorted by age from lowest to highest.

5. Group By using $group

MongoDB does not use SQL-style **GROUP BY**. Instead, use the aggregation pipeline with the **$group** stage.

Practice the following questions:

Q14. Count how many students are in each course (BCA and BSc).

Q15. Find the average marks for each course.

Q16. Find the maximum marks for each course.

Q17. Find the minimum marks for each course.

Q18. Count how many students belong to each city.

Q19. Find the average marks for each city.

6. Challenge Questions

Q20. Find the course having the highest average marks.

Q21. Find cities having more than 1 student.

Q22. Find the total number of students in the collection.

Useful MongoDB Syntax Reference

Task	Syntax
Insert one	db.students.insertOne({ ... })
Insert many	db.students.insertMany([{ ... }, { ... }])
Find all	db.students.find()
Find with condition	db.students.find({ course: "BCA" })
Update one	db.students.updateOne({ name: "Priya" }, { $set: { marks: 90 } })
Delete one	db.students.deleteOne({ name: "Karan" })
Sort	db.students.find().sort({ marks: -1 })
Limit	db.students.find().limit(3)
Group	db.students.aggregate([{ $group: { _id: "$course", count: { $sum: 1 } } }])

Submission Checklist

■ Create collegeDB and students collection.

■ Insert all sample records.

■ Complete Q1–Q22.

■ Run every query in MongoDB Compass or mongosh.

■ Save screenshots of your output for verification.